

Olympiad Test Paper — Mathematics (Class 7) — Paper 3

Chapter 6: Number Play

Paper 3 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	6 — Number Play
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. An odd count of odd numbers gives an odd sum; the Virahanka-Fibonacci rule adds the two previous terms (never a constant or a ratio); in the taller-children game the first child always says 0; Kaprekar's routine always lands on 6174 and Collatz always falls to 1. Do not write on this paper — mark answers on your answer sheet.

1. In the taller-children-in-front game, every child calls how many taller children stand ahead of them. A line of 6 children calls out, front to back: 0, 1, 3, 0, 2, 1. Exactly one of these calls is impossible no matter how the heights are arranged. Which one?

- (A) the 3rd child, who said 3 (B) the 1st child, who said 0
(C) the 5th child, who said 2 (D) the 6th child, who said 1

2. Kishor adds up some odd numbers and some even numbers. He used 5 odd numbers and 8 even numbers. What is the parity of his total?

- (A) even (B) odd
(C) it depends on the actual even numbers (D) it depends on the actual odd numbers

3. A Virahanka-Fibonacci-type sequence (each term is the sum of the two before it) has its 4th term equal to 13 and its 6th term equal to 34. What is its 1st term?

- (A) 3 (B) 2
(C) 5 (D) 1

4. In the cryptarithm $AB + AB + AB = CAB$ (A and B are the digits of a two-digit number, C is a single digit, and the three-digit answer ends in the same two digits AB), what is the two-digit number AB?

(A) 25

(B) 75

(C) 10

(D) 50

5. In a 3×3 grid a 'supercell' is a cell strictly larger than every neighbour sharing an edge with it. If you fill the grid with 1-9 (each once) to get as MANY supercells as possible, what is the greatest number of supercells achievable?

(A) 4

(B) 3

(C) 5

(D) 9

6. A 3×3 magic square uses 1-9 once each so every row, column and main diagonal has the same sum. A student fills the centre with 4 instead of 5. Why can no valid magic square result?

(A) the centre must be the largest number, 9

(B) the magic sum must be 15 and the centre must be 15 divided by 3 = 5, so the centre cannot be 4

(C) 4 is even and the centre must be odd by the row rule

(D) the centre may be any digit; some other cell is the real problem

7. In the Collatz ($3n+1$) process you replace an even number by half of it and an odd number n by $3n+1$, repeating until you reach 1. Starting from 6, how many steps does it take to first reach 1?

(A) 7

(B) 8

(C) 9

(D) 6

8. Kaprekar's routine on a 4-digit number (with at least two different digits): arrange the digits in descending then ascending order, subtract, repeat. Whatever you start with, the process eventually reaches one fixed number. What is it?

(A) 1089

(B) 9999

(C) 0

(D) 6174

9. A line of 5 children plays taller-in-front and the calls front-to-back are 0, 2, 1, 0, 1. Why is this sequence impossible?

(A) the 1st child should not say 0

(B) the 2nd child claims 2 taller children ahead, but only 1 child stands ahead of the 2nd position

(C) no child may ever say a number larger than 1

(D) the calls must always increase from front to back

10. How many different rhythms of exactly 8 beats can be formed using short (1-beat) and long (2-beat) syllables, where order matters?

(A) 21

(B) 34

(C) 55

(D) 32

11. How many three-digit palindromes are divisible by 3?

(A) 33

(B) 30

(C) 27

(D) 90

12. A Virahanka-Fibonacci sequence starts 1, 1, 2, 3, 5, 8, 13, 21, 34, What is the units digit of the 15th term?

(A) 0

(B) 7

(C) 5

(D) 1

13. What is the smallest 4-digit number that is divisible by 9 and reads the same forwards and backwards (a palindrome)?

(A) 1001

(B) 1111

(C) 1881

(D) 1818

14. The cryptarithm $ABA \times 4 = BAB$ has no solution, so instead consider $AB + BA = 99$, where AB and BA are both genuine two-digit numbers (so neither A nor B is 0). If A and B are digits with $A > B$, how many different ordered pairs (A, B) are possible?

(A) 9

(B) 5

(C) 4

(D) 8

15. A 3×3 magic square of 1-9 has magic sum 15. The four corner cells are filled with even numbers. What number must sit in the centre?

(A) 4

(B) 6

(C) 9

(D) 5

16. Starting the Collatz process from 7, what is the largest number reached before the sequence falls to 1?

(A) 40

(B) 26

(C) 52

(D) 22

17. A line of taller-in-front children gives the calls 0, 1, 2, 3, 4, 5 front to back. How many children are taller than the very FIRST child?

(A) 5

(B) 0

(C) 1

(D) cannot be determined

18. Kaprekar's routine for THREE-digit numbers (descending minus ascending, repeat) settles on a single fixed value for any starting number that is not a repdigit. What is that value?

(A) 6174

(B) 495

(C) 999

(D) 0

19. In a 3×3 grid filled with 1-9 (no repeats), what is the SMALLEST possible number of supercells (a supercell is strictly greater than all its edge-neighbours)?

(A) 1

(B) 0

(C) 2

(D) 4

20. The Virahanka-Fibonacci sequence 1, 1, 2, 3, 5, 8, 13, 21, 34, 55 has the property that the sum of the first n terms equals (the $(n+2)$ th term) minus 1. Use this to find $1 + 1 + 2 + 3 + 5 + 8 + 13 + 21 + 34$.

- (A) 89 (B) 87
(C) 143 (D) 88

21. What is the largest 3-digit number using three DIFFERENT non-zero digits whose digit-sum is exactly 6?

- (A) 411 (B) 312
(C) 321 (D) 123

22. Kishor writes 9 numbers. He says their sum is even. He also says exactly 4 of them are odd. Is this consistent?

- (A) Yes - an even count of odd numbers gives an even sum, and the 5 even numbers keep it even
(B) No - 9 numbers always sum to an odd total
(C) Yes, but only if the odd numbers are all equal
(D) No - 4 odd numbers always make an odd sum

23. What is the sum of the largest and the smallest 4-digit palindromes?

- (A) 10890 (B) 11000
(C) 10989 (D) 9999

24. In a 3×3 grid of 1-9 (no repeats), the top row sums to 8 and the left column sums to 11. The two share the top-left corner cell, which holds 1. What do the top row's OTHER two cells and left column's other two cells sum to together?

- (A) 19 (B) 18
(C) 16 (D) 17

25. The Collatz process is run from 1. What does the sequence do?

- (A) it stops immediately and never moves (B) it cycles 1, 4, 2, 1, 4, 2, ... forever
(C) it grows without bound (D) it reaches 0

26. A two-digit number plus its reverse equals 143, AND the two digits differ by 1. What is the original (larger) number?

- (A) 94 (B) 85
(C) 67 (D) 76

27. In the cryptarithm $AB \times 3 = CCB$ (AB a two-digit number; CCB a three-digit number whose first two digits are equal and whose units digit B equals the units digit of AB), find AB .

- (A) 75 (B) 85
(C) 45 (D) 37

- 28.** The first ten Virahanka-Fibonacci terms are 1,1,2,3,5,8,13,21,34,55. Among these, how many are EVEN?
- (A) 3 (B) 4
(C) 2 (D) 5
- 29.** A 3-digit palindrome is also a multiple of 11. How many such numbers are there?
- (A) 8 (B) 9
(C) 10 (D) 90
- 30.** Run the Collatz process from 11. How many steps does it take to reach 1?
- (A) 13 (B) 15
(C) 14 (D) 12
- 31.** In a 3x3 grid of 1-9 (no repeats), the centre cell is part of how many of the eight 'lines' (3 rows, 3 columns, 2 diagonals)?
- (A) 4 (B) 2
(C) 3 (D) 8
- 32.** Five DISTINCT odd numbers are added. Which of the following can the total NOT be?
- (A) 35 (B) 49
(C) 40 (D) 63
- 33.** What is the smallest two-digit number n such that the Collatz process from n takes MORE THAN 10 steps to reach 1?
- (A) 12 (B) 11
(C) 10 (D) 14
- 34.** A magic square of 1-9 has magic sum 15. One diagonal reads 2, 5, 8 from top-left to bottom-right. What is the SUM of the other two corners (top-right and bottom-left)?
- (A) 10 (B) 12
(C) 15 (D) 8
- 35.** In the taller-in-front game, a line is arranged so the calls are o, o, o, o. What does this say about the heights?
- (A) all four children are exactly the same height
(B) the children are arranged tallest at the front
(C) the calls are impossible for four children
(D) each child is taller than everyone ahead of them, i.e. heights increase from front to back
- 36.** The number 1729 is run through Kaprekar's 4-digit routine. After the FIRST subtraction step, what number do you get? (Use digits 1,7,2,9.)
- (A) 6174 (B) 8262
(C) 7443 (D) 8442

37. A 3-digit number has digit sum 18 and is a palindrome. How many such numbers are there?

- (A) 4 (B) 6
(C) 9 (D) 5

38. Two consecutive terms of a Virahanka-Fibonacci-type sequence (each term = sum of the two before) are 50 and 81. What is the term TWO places BEFORE the 50?

- (A) 31 (B) 12
(C) 19 (D) 13

39. In a 3x3 grid of 1-9 (no repeats), the four corner cells sum to 20 and the four edge-midpoint cells sum to 20. What number is in the centre?

- (A) 4 (B) 6
(C) 5 (D) 9

40. Which of these four-digit numbers is divisible by BOTH 9 and 11?

- (A) 8901 (B) 8910
(C) 8190 (D) 9810

41. A line of 7 children plays taller-in-front. The maximum number any single child could possibly call out is which value?

- (A) 7 (B) 5
(C) 0 (D) 6

42. The Virahanka-Fibonacci sum identity says $1 + 1 + 2 + \dots + F_n = F_{(n+2)} - 1$. If the sum of the first n terms is 142, what is $F_{(n+2)}$?

- (A) 144 (B) 143
(C) 142 (D) 141

43. In the cryptarithm $AB - BA = 27$ (AB and BA two-digit numbers, $A > B$), what is $A - B$?

- (A) 3 (B) 2
(C) 9 (D) 27

44. Among the numbers 1 to 100, how many are unchanged when you run a SINGLE step of Kaprekar's 3-digit routine after padding to three digits with leading zeros? (Consider only the genuine 3-digit Kaprekar fixed point.)

- (A) all repdigits like 11, 22 are fixed (B) 100 is fixed
(C) every multiple of 9 is fixed
(D) the only 3-digit Kaprekar fixed value is 495, so none of 1 to 100 is fixed

45. A 3x3 magic square of 1-9 (magic sum 15) is reflected left-to-right (each row reversed). Is the result still a magic square?

- (A) yes - rows keep their sums, columns are just reordered, and the diagonals swap roles but each still sums to 15
(B) no - the diagonals are destroyed (C) no - reflection changes the centre value

(D) only if the original used the numbers 1 to 9

46. Starting Collatz from 27 is famous for taking many steps. Without computing all of them, what can you be CERTAIN of about where it ends?

(A) it grows forever and never comes back down

(B) it settles on 6174

(C) it must eventually reach 1 (every tested starting number does, including 27)

(D) it ends at the largest number it reaches

47. What is the SMALLEST two-digit number that equals 7 times the sum of its own digits?

(A) 42

(B) 63

(C) 21

(D) 14

48. In a 3x3 grid of 1-9 (no repeats), what is the LARGEST possible value of (sum of one chosen row) given the constraint that the same grid's main diagonal must contain the number 1?

(A) 23

(B) 22

(C) 24

(D) 15

49. The Virahanka-Fibonacci sequence's terms grow so that each term is roughly 1.618 times the one before. Using this, about how many times bigger is the 10th term than the 7th term?

(A) about 1.6 times

(B) about 4.2 times

(C) about 3 times

(D) about 8 times

50. A palindrome year like 2002 reads the same forwards and backwards. How many palindrome years are there from 2000 to 2999 inclusive?

(A) 9

(B) 11

(C) 10

(D) 100

51. Six children play taller-in-front and call 0, 1, 1, 2, 3, 5. Notice these are Virahanka-Fibonacci-like numbers, but is the SEQUENCE OF CALLS valid for the game?

(A) no - calls can never repeat the value 1

(B) yes - each call k is at most (position - 1), and 0,1,1,2,3,5 each obey that bound

(C) no - the last call 5 is too large

(D) only if the children's heights also follow the Fibonacci pattern

52. What is the sum of all 3-digit palindromes whose middle digit is 0 (like 101, 202, ...)?

(A) 4545

(B) 4995

(C) 5050

(D) 4950

53. A 3x3 magic square of 1-9 has magic sum 15. If you ADD 10 to every cell, what is the new magic sum?

- | | |
|---------|--------|
| (A) 25 | (B) 15 |
| (C) 150 | (D) 45 |

54. Run Collatz from 16. How many steps to reach 1, and what kind of number is 16 that makes this short?

- | | |
|--|--|
| (A) 5 steps, because 16 is even | (B) 4 steps, because 16 is a multiple of 4 |
| (C) 16 steps, one per unit | |
| (D) 4 steps, because 16 is a power of 2 so it just halves repeatedly | |

55. A number n and its digit-reversal differ by 0. What can be said about n if n is a 3-digit number?

- | | |
|---|-----------------------|
| (A) n must be a multiple of 11 | (B) n must end in 0 |
| (C) n must be a single-digit number | |
| (D) n is a palindrome (its first and last digits are equal) | |

56. The first child in a taller-in-front line of n children always calls 0. For a line of any size, how many children are GUARANTEED to be unable to call a number larger than 0?

- | | |
|---|--------------------------------------|
| (A) all of them | (B) exactly 2 - the first and second |
| (C) exactly 1 - only the first child is forced to 0 | |
| (D) none - even the first child may call more | |

57. In a 3×3 grid of 1-9 (no repeats), you want EXACTLY 2 supercells (each strictly greater than all edge-neighbours). One supercell must be the cell holding 9. Where can the second supercell holding 8 be placed so it is NOT adjacent to 9's cell?

- | |
|---|
| (A) in a cell sharing no edge with 9's cell, e.g. the opposite corner |
| (B) directly beside 9 in the same row |
| (C) in the centre, which touches every other cell |
| (D) nowhere - 8 can never be a supercell |

58. The largest 3-digit number that stays a 3-digit number after one Collatz step from it being even is which? (You halve an even 3-digit number; the result should still have 3 digits.)

- | | |
|---------|--|
| (A) 998 | (B) 999 |
| (C) 500 | (D) 998 is even but 999 is the true answer |

59. A Virahanka-Fibonacci-type sequence has 8th term 21 and 10th term 55. What is the 9th term?

- | | |
|--------|--------|
| (A) 33 | (B) 38 |
| (C) 29 | (D) 34 |

60. Among numbers 100 to 999, a palindrome aba is even exactly when its first digit a is even. How many even 3-digit palindromes are there?

- | | |
|--------|--------|
| (A) 45 | (B) 50 |
| (C) 40 | (D) 20 |

Solutions — Mathematics (Class 7), Chapter 6:

Number Play — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	B	21	C	31	A	41	D	51	B
2	B	12	A	22	A	32	C	42	B	52	A
3	A	13	C	23	B	33	B	43	A	53	D
4	D	14	C	24	D	34	A	44	D	54	D
5	C	15	D	25	B	35	D	45	A	55	D
6	B	16	C	26	D	36	D	46	C	56	C
7	B	17	B	27	A	37	D	47	C	57	A
8	D	18	B	28	A	38	C	48	C	58	A
9	B	19	A	29	A	39	C	49	B	59	D
10	B	20	D	30	C	40	B	50	C	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) A child can only count taller children standing AHEAD, so the call at position p (counting from the front, the first child being position 1) can be at most $p - 1$. Check each call against this bound: the 1st child (0 ahead) can say at most 0, and says 0 - fine; the 2nd (1 ahead) can say at most 1, says 1 - fine; the 3rd child has only 2 people ahead, so the most it could ever say is 2, yet it says 3 - impossible; the 4th (3 ahead) says 0 - fine; the 5th (4 ahead) says 2 - fine; the 6th (5 ahead) says 1 - fine. Only the 3rd child's call of 3 breaks the bound, so that call is the impossible one. The first child's 0 is always correct, and the 5th and 6th calls are well within their limits.

2. (B) Even numbers contribute nothing to parity - any count of them sums to an even amount. So the total's parity equals the parity of the sum of the 5 odd numbers. An odd COUNT of odd numbers gives an odd sum (each odd is 1 more than an even, and 5 such 'extra ones' add to an odd surplus). Hence the total is odd, regardless of the actual sizes. The 'even' answer wrongly assumes any mix balances out; the 'depends on the even/odd

values' answers ignore that parity is fixed purely by the COUNT of odds, not their magnitudes.

3. (A) Call the first two terms a and b . Then the 3rd term is $a+b$, the 4th is $a+2b$, the 5th is $2a+3b$ and the 6th is $3a+5b$. We are told $a+2b = 13$ and $3a+5b = 34$. From the first, $a = 13 - 2b$. Substitute into the second: $3(13 - 2b) + 5b = 39 - 6b + 5b = 39 - b = 34$, so $b = 5$ and $a = 13 - 2(5) = 3$. Check the 6th term: $3(3) + 5(5) = 9 + 25 = 34$ ✓. So the 1st term is 3. The value 2 comes from misreading which term equals 13; 5 confuses the first term with b ; 1 is the default Fibonacci start that does not fit the data.

4. (D) Adding AB three times gives $3 \times AB$. We need $3 \times AB$ to be a three-digit number whose last two digits are AB itself, i.e. $3 \times AB = 100C + AB$, so $2 \times AB = 100C$, $AB = 50C$. For AB to be two digits, $C = 1$ gives $AB = 50$ and $3 \times 50 = 150 = C A B$ with $C=1, A=5, B=0$ ✓. $C = 2$ would give $AB = 100$, too big. So $AB = 50$. The value 25 gives 75 (not ending in 25); 75 gives 225 (ends in 25, not 75); 10 gives 30 (not three digits).

5. (C) Two edge-adjacent cells can never both be supercells, since each would have to exceed the other. So the supercells form an 'independent set': no two share an edge. On the 3×3 grid the four corners and the centre are pairwise non-edge-adjacent (each corner touches only edge-midpoints, and the centre also touches only edge-midpoints), so these 5 cells form a checkerboard pattern. Place the four SMALLEST values 1,2,3,4 on the four edge-midpoints and the five largest 5,6,7,8,9 on the corners and centre: then every corner (≥ 6) beats its two edge-midpoint neighbours (all ≤ 4), and the centre (5) beats its four edge-midpoint neighbours (1,2,3,4). All 5 chosen cells are supercells at once. No arrangement beats 5, because a 6th supercell would have to be an edge-midpoint, which is edge-adjacent to two corners that are already larger. So the maximum is 5. Answering 4 stops one short (it forgets the centre joins the four corners as a non-adjacent fifth); 3 undercounts; 9 is impossible since edge-adjacent cells block each other.

6. (B) The total $1+2+\dots+9 = 45$ splits into 3 equal rows, so the magic sum is $45/3 = 15$. The centre cell lies on 4 lines (one row, one column, two diagonals). Summing those 4 lines counts the centre 4 times and every other cell once; that total is $4 \times 15 = 60$, while the eight non-centre cells sum to $45 - \text{centre}$. So $60 = (45 - \text{centre}) + 4 \times \text{centre} = 45 + 3 \times \text{centre}$, giving $3 \times \text{centre} = 15$, $\text{centre} = 5$. A centre of 4 is therefore impossible. The 'centre must be 9' answer confuses largest with central; the 'must be odd' answer states a true fact (centre is 5, odd) but for the wrong reason; the 'any digit' answer is simply false.

7. (B) Apply the rule from 6: 6 is even $\rightarrow 3$; 3 is odd $\rightarrow 3(3)+1 = 10$; 10 $\rightarrow 5$; 5 is odd $\rightarrow 16$; 16 $\rightarrow 8$; 8 $\rightarrow 4$; 4 $\rightarrow 2$; 2 $\rightarrow 1$. Counting the arrows: $6 \rightarrow 3 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$ is 8 steps. The answer 7 forgets to count the final $2 \rightarrow 1$; 9 double-counts a step; 6 stops early at 4 or 2.

8. (D) Kaprekar's constant for four-digit numbers is 6174: from $7641 - 1467 = 6174$, the routine reproduces 6174 forever, and every starting four-digit number (not all-equal digits)

reaches it within at most 7 steps. 1089 is the famous three-digit-reverse-product curiosity, not the Kaprekar fixed point; 9999 has all equal digits and gives 0, which is the degenerate case the rule excludes; 0 is only reached when all four digits are equal, which is barred. So the universal landing value is 6174.

9. (B) A child's call counts taller children standing AHEAD, so the call at position p (front child = position 1) can be at most $p - 1$. The 2nd child has only 1 person ahead, so the largest number it could possibly say is 1, yet it says 2 - more taller children than there are people in front of it. That is impossible regardless of heights, so the sequence fails right there. The first child saying 0 is always correct (nobody is ahead); calls can certainly exceed 1 further back (e.g. a 4th child could say 3); and calls need not increase - patterns like 0,1,0 are perfectly valid.

10. (B) The count of beat-patterns of length n follows the Virahanka-Fibonacci rule: a pattern either starts with a short (leaving $n-1$ beats) or a long (leaving $n-2$ beats), so $C(n) = C(n-1) + C(n-2)$. With $C(1)=1$, $C(2)=2$, we get $C(3)=3$, $C(4)=5$, $C(5)=8$, $C(6)=13$, $C(7)=21$, $C(8)=34$. So 34 rhythms. The value 21 is $C(7)$ (off by one beat); 55 is $C(9)$ (one too many); $32 = 2^5$ wrongly treats every beat as a free binary choice, ignoring that longs use two beats.

11. (B) A three-digit palindrome has the form aba with a from 1-9 and b from 0-9; there are $9 \times 10 = 90$ of them. Its digit sum is $2a + b$, and it is divisible by 3 exactly when $2a + b$ is a multiple of 3. For each fixed a (9 choices), b ranges over 0-9 (10 values); among any 10 consecutive integers the residues mod 3 are not perfectly even, but $2a+b$ runs through 10 consecutive values as b goes 0-9, of which either 3 or 4 are multiples of 3. Counting precisely: b in $\{0..9\}$ gives multiples of 3 at b values making $2a+b$ divisible by 3 - exactly 3 or 4 per a . Summing over all a , the multiples of 3 among palindromes total $90/3 = 30$, since the map is uniform across the full set (the palindromes aba in order 101,111,...,999 hit every residue class equally often: exactly one third). Hence 30. The value 90 is the count of ALL palindromes; 33 and 27 come from miscounting b -values per a .

12. (A) List terms (term 1 = 1): 1,1,2,3,5,8,13,21,34,55,89,144,233,377,610. The 15th term is 610, whose units digit is 0. Tracking only units digits also works:

1,1,2,3,5,8,3,1,4,5,9,4,3,7,0 - the 15th units digit is 0. The value 7 is the units digit of the 14th term (377); 5 is the 10th term's units digit (55); 1 is the units digit of several early terms but not the 15th.

13. (C) A 4-digit palindrome has the form $abba$ with digit sum $2a + 2b = 2(a+b)$. It is divisible by 9 when $2(a+b)$ is a multiple of 9, i.e. $a+b$ is a multiple of 9 (since 2 and 9 share no factor). To be smallest, take $a = 1$ (smallest leading digit), so b must make $a+b$ a multiple of 9: $a+b = 9$ gives $b = 8$. The number is 1 8 8 1 = 1881, and $1+8+8+1 = 18$ is divisible by 9. The value 1001 has digit sum 2 (not divisible by 9); 1111 has digit sum 4; 1818 is not a palindrome (reversed it is 8181).

14. (C) Writing the two-digit numbers in place value, $AB + BA = (10A + B) + (10B + A) = 11(A + B) = 99$, so $A + B = 9$. Both AB and BA must be real two-digit numbers, so $A \geq 1$ and $B \geq 1$ (no leading zero); thus B cannot be 0. With $A > B$ and $A + B = 9$ and $B \geq 1$, the pairs are $(8,1)$, $(7,2)$, $(6,3)$ and $(5,4)$ -- exactly 4. Counting $(9,0)$ as well gives 5, but BA would be '09', not a two-digit number, so it is excluded. Answering 9 confuses the digit count with the sum; 8 double-counts by allowing $B > A$ as well.

15. (D) In any 3×3 magic square of 1-9 the centre is forced to 5 (the magic sum is $45/3 = 15$, and the centre, lying on 4 lines, must equal $15/3 = 5$ by the line-counting argument). The corners must be the four even numbers 2,4,6,8 and the edge-midpoints the remaining odds 1,3,7,9 - this is exactly the structure of the standard magic square. So whatever placement of the even corners, the centre is 5. The values 4, 6 are corner numbers (even), not the centre; 9 is an edge number here, never the centre.

16. (C) From 7: 7 is odd $\rightarrow 22$; $22 \rightarrow 11$; $11 \rightarrow 34$; $34 \rightarrow 17$; $17 \rightarrow 52$; $52 \rightarrow 26$; $26 \rightarrow 13$; $13 \rightarrow 40$; $40 \rightarrow 20 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$. The peak values along the way are 22, 34, 52, 40 - the maximum is 52. The value 40 is a later local high but smaller than 52; 26 and 22 are intermediate values, not the maximum.

17. (B) The first child always says 0, meaning no taller child stands ahead - trivially true since nobody is ahead. But the question asks how many taller children there are anywhere relative to the first child. With calls 0,1,2,3,4,5 strictly increasing, each successive child has one more taller person ahead, which is the pattern when the line is arranged from tallest at front to shortest at back: the front child is the TALLEST of all, so zero children are taller than the first child. The value 5 confuses the back child's call with the first child; 1 misreads the second child's call; it is fully determined, so 'cannot be determined' is wrong.

18. (B) For three-digit numbers Kaprekar's constant is 495: e.g. from $954 - 459 = 495$, and $954-459$ reproduces 495. Every three-digit number with at least two distinct digits reaches 495 within a few steps. 6174 is the FOUR-digit Kaprekar constant - the classic trap of using the wrong width; 999 is a repdigit which gives 0; 0 only occurs for repdigits, which the rule excludes.

19. (A) There must always be at least one supercell: the cell holding 9, the largest number, exceeds all of its neighbours (nothing is bigger), so 9's cell is always a supercell. Hence 0 is impossible. Can we hold the count to exactly 1? Place 9 in the centre so it neighbours 4 cells; arrange the rest so that no other cell beats all its neighbours - e.g. make the grid increase toward the centre along every path so only the centre is a local maximum. This is achievable, giving exactly 1 supercell. The value 0 ignores that 9 is always a supercell; 2 and 4 overstate the forced minimum.

20. (D) There are 9 terms here (up to 34, the 9th term). By the identity, the sum of the first 9 terms equals the 11th term minus 1. The sequence is 1,1,2,3,5,8,13,21,34,55,89, so the 11th term is 89, and the sum is $89 - 1 = 88$. Direct addition confirms:

$1+1+2+3+5+8+13+21+34 = 88$. The value 89 forgets to subtract 1 (it is the 11th term itself); 87 over-subtracts; 143 mistakenly adds the 10th term 55 as well.

21. (C) We need three different non-zero digits adding to 6; the only such set is $\{1, 2, 3\}$ (since $1+2+3 = 6$ and any other distinct non-zero triple exceeds 6). To make the largest number, place the digits in descending order: 3, 2, 1 $\rightarrow$ 321. The value 411 repeats the digit 1 and uses a zero-free but non-distinct set; 312 is a valid arrangement of 3,1,2 but smaller than 321; 123 is the smallest arrangement.

22. (A) Parity of a sum depends only on the count of odd addends. Here 4 odds are present; an even count of odd numbers sums to an even amount, and the remaining 5 even numbers add nothing to parity. So the total is even - consistent. The claim that 9 numbers always sum odd is false (parity depends on how many are odd, not the total count); the equality condition is irrelevant to parity; and 4 odds give an EVEN, not odd, sum - that distractor inverts the rule.

23. (B) A 4-digit palindrome has the form abba with leading digit a from 1-9 and inner digit b from 0-9. The smallest takes the smallest leading digit and smallest inner digit: a = 1, b = 0 gives 1001. The largest takes a = 9, b = 9 giving 9999. Their sum is $1001 + 9999 = 11000$. The value 10890 mistakenly treats 0990 as the smallest (which is not a 4-digit number); 10989 is an addition slip; 9999 forgets to add the smallest palindrome at all.

24. (D) Top row sums to 8 and includes the corner cell (1), so its other two cells sum to $8 - 1 = 7$. Left column sums to 11 and includes the same corner cell (1), so its other two cells sum to $11 - 1 = 10$. The 'other' cells of the row and the 'other' cells of the column are four distinct cells (none is the shared corner), so together they sum to $7 + 10 = 17$. The value 19 forgets to remove the corner from BOTH lines; 18 removes it once; 16 subtracts an extra 1.

25. (B) At 1 (odd) the rule $3n+1$ gives 4; $4 \rightarrow 2$; $2 \rightarrow 1$; and the loop repeats: $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$. So once you reach 1, the process enters the cycle 1,4,2 indefinitely. It does not freeze at 1 (the rule still applies to 1), it does not grow forever, and it never reaches 0 (the rule keeps numbers positive). This 1-4-2 loop is exactly why 1 is treated as the stopping point in the puzzle.

26. (D) Let the number be $10t + u$; its reverse is $10u + t$, and their sum is $11(t + u) = 143$, so $t + u = 13$. The extra condition that the digits differ by 1 means $t - u = 1$ (taking t larger). Solving $t + u = 13$ and $t - u = 1$ gives $t = 7$, $u = 6$, so the larger number is 76 (and $76 + 67 = 143$ ✓). The pair (9,4) gives 94 but the digits differ by 5, not 1; (8,5) gives 85 with a difference of 3; 67 is the reverse (smaller) number, not the original larger one.

27. (A) The result must end in B, the units digit of AB, so $3 \times B$ must end in B, meaning $2B$ is a multiple of 10, giving $B = 0$ or $B = 5$. Try $B = 5$ with various tens digits: $35 \times 3 = 105$ (digits 1,0,5 - not CCB), $45 \times 3 = 135$ (no), $65 \times 3 = 195$ (no), $75 \times 3 = 225$ (digits 2,2,5 - that IS CCB with $C = 2$ and $B = 5$, and units 5 matches the units of 75 ✓). Check $85 \times 3 = 255$:

digits 2,5,5 - the first two differ, so 85 fails. So the unique answer is 75. The value 85 gives 255 (leading digits unequal); 45 gives 135 (not CCB); 37 gives 111 whose units 1 does not match the units of 37.

28. (A) Check each:

1(odd),1(odd),2(even),3(odd),5(odd),8(even),13(odd),21(odd),34(even),55(odd). The even terms are 2, 8, 34 - that is 3 of them. The pattern of parity is O,O,E repeating, so every third term is even; among 10 terms positions 3, 6, 9 are even, giving 3. The value 4 over-counts; 2 misses one; 5 confuses odds with evens (there are 7 odds).

29. (A) A 3-digit palindrome aba is a multiple of 11 when its alternating digit sum $a - b + a = 2a - b$ is a multiple of 11. With a from 1-9 and b from 0-9, $2a - b$ ranges from -9 to 18, so the only multiples of 11 in range are 0 and 11. For $2a - b = 0$: $b = 2a$, valid for $a=1$ (121), $a=2$ (242), $a=3$ (363), $a=4$ (484) - 4 numbers ($a=5$ needs $b=10$, invalid). For $2a - b = 11$: $b = 2a - 11$, valid for $a=6$ (616), $a=7$ (737), $a=8$ (858), $a=9$ (979) - 4 numbers ($a=5$ gives $b=-1$, invalid). Total $4 + 4 = 8$. The value 9 over-counts by one near-miss; 10 over-counts further; 90 is the count of ALL 3-digit palindromes, not just the multiples of 11.

30. (C) From 11: 11->34->17->52->26->13->40->20->10->5->16->8->4->2->1. Count the arrows: 11->34 (1), ->17 (2), ->52 (3), ->26 (4), ->13 (5), ->40 (6), ->20 (7), ->10 (8), ->5 (9), ->16 (10), ->8 (11), ->4 (12), ->2 (13), ->1 (14). So 14 steps. The value 13 forgets the final 2->1; 15 adds a phantom step; 12 miscounts the descent.

31. (A) The centre lies on the middle row, the middle column, and BOTH main diagonals - that is $1 + 1 + 2 = 4$ lines. This is exactly why the centre is so constrained in a magic square. A corner lies on 3 lines (its row, column, one diagonal); an edge-midpoint lies on only 2. The value 2 is the edge-cell count; 3 is the corner count; 8 is the total number of lines, not how many touch the centre.

32. (C) Five odd numbers - an odd count of odds - always sum to an ODD total. So any even total is impossible. Among the choices, 40 is even, hence unreachable; 35, 49 and 63 are all odd and achievable (e.g. $1+3+5+7+19 = 35$, etc.). The trap is to check sizes rather than parity: an even target like 40 is ruled out instantly by the odd-count-of-odds rule.

33. (B) Count steps for the smallest two-digit numbers. From 10: 10->5->16->8->4->2->1 is 6 steps - not enough. From 11: 11->34->17->52->26->13->40->20->10->5->16->8->4->2->1 is 14 steps - more than 10, and 11 is the smallest two-digit value, so it qualifies first. For comparison 12->6->3->10->5->16->8->4->2->1 is only 9 steps (fails the >10 test), and 14->7->... takes 17 steps but 14 is larger than 11. So the smallest qualifying two-digit number is 11. The value 12 actually falls short at 9 steps; 10 takes only 6; 14 qualifies but is not the smallest.

34. (A) The other two corners are the ends of the anti-diagonal, which also passes through the centre 5. Since every line (including that diagonal) sums to 15, we have top-right + 5 +

bottom-left = 15, so the two corners sum to $15 - 5 = 10$. (Concretely they are the remaining even numbers 4 and 6, which indeed add to 10.) The value 12 wrongly adds 4 and 8; 15 forgets to remove the centre; 8 picks the wrong pair.

35. (D) A call of 0 means no taller child stands ahead. For every child to say 0, each child must be at least as tall as all those in front - with distinct heights this means each child is taller than everyone ahead, so heights strictly increase front to back (shortest at front, tallest at back). It does NOT mean equal heights (that is one degenerate way but the game assumes a clear order); 'tallest at front' would give increasing calls 0,1,2,3, not all zeros; and the all-zero pattern is perfectly possible, so it is not impossible.

36. (D) Arrange the digits 1, 7, 2, 9 in descending order: 9721. Arrange them in ascending order: 1279. Subtract: $9721 - 1279 = 8442$. So the first step gives 8442. The value 6174 is Kaprekar's eventual fixed point, reached only after several more steps, not on the first; 8262 and 7443 are arithmetic slips in the subtraction.

37. (D) A 3-digit palindrome aba has digit sum $2a + b = 18$ with a in 1-9 and b in 0-9. Solve $b = 18 - 2a$, requiring $0 \leq b \leq 9$: $18 - 2a \geq 0$ means $a \leq 9$, and $18 - 2a \leq 9$ means $2a \geq 9$, $a \geq 4.5$, so $a \geq 5$. Thus a in $\{5,6,7,8,9\}$: $a=5, b=8$ (585); $a=6, b=6$ (666); $a=7, b=4$ (747); $a=8, b=2$ (828); $a=9, b=0$ (909). That is 5 numbers. The value 4 misses one endpoint; 6 includes an out-of-range b ; 9 ignores the $b \leq 9$ constraint.

38. (C) If two consecutive terms are P and Q with P before Q , the term just before P is $Q - P$ (since $P + (\text{the one before } P)$ would not help - rather, the term before P equals (the term after) minus P only forward). Work backward using term = next - previous: the term before 50 is $81 - 50 = 31$. The term before that 31 is $50 - 31 = 19$. So two places before 50 is 19. The value 31 is only ONE place before 50; 12 and 13 come from subtracting in the wrong order or one step too few.

39. (C) The nine numbers 1-9 sum to 45. The four corners (20) plus the four edges (20) plus the centre = 45, so centre = $45 - 20 - 20 = 5$. This matches the magic-square fact that 5 sits centre. The values 4, 6, 9 each fail the total $45 = 20 + 20 + \text{centre}$.

40. (B) Divisible by 9 needs the digit sum to be a multiple of 9; divisible by 11 needs the alternating digit sum to be a multiple of 11. Check 8910: digit sum $8+9+1+0 = 18$ (multiple of 9 ✓); alternating sum $8 - 9 + 1 - 0 = 0$ (multiple of 11 ✓). So 8910 works ($8910 = 99 \times 90$). Check 8901: digit sum 18 ✓ for 9, but alternating $8 - 9 + 0 - 1 = -2$, not a multiple of 11. Check 8190: digit sum 18 ✓ for 9, but alternating $8 - 1 + 9 - 0 = 16$, not a multiple of 11. Check 9810: digit sum 18 ✓ for 9, but alternating $9 - 8 + 1 - 0 = 2$, not a multiple of 11. Only 8910 passes both tests.

41. (D) A child can only see children AHEAD, so the most taller children possible is everyone in front. The back (7th) child has 6 people ahead, so the largest call any child can make is 6 (only the back child can reach it, by being the shortest with all 6 ahead taller).

No child can say 7 (there are not 7 people ahead of anyone in a 7-line); 5 understates the maximum; 0 is the minimum (the front child's call).

42. (B) By the identity, sum of first n terms = $F(n+2) - 1$. If that sum is 142, then $F(n+2) - 1 = 142$, so $F(n+2) = 143$. The value 144 forgets to add only 1 (it would need the sum to be 143); 142 omits the +1 entirely; 141 subtracts instead of adding.

43. (A) $AB - BA = (10A + B) - (10B + A) = 9A - 9B = 9(A - B)$. Set $9(A - B) = 27$, so $A - B = 3$. The value 2 comes from dividing 27 by a wrong factor; 9 confuses the multiplier with the answer; 27 ignores the factor of 9 entirely. (For instance $A=5, B=2$ gives $52-25 = 27$ with $A-B = 3$.)

44. (D) Kaprekar's 3-digit routine has exactly one non-trivial fixed point: 495 (since $954 - 459 = 495$ repeats). No number from 1 to 100 equals 495, so none of them is a true Kaprekar fixed point. Repdigits like 11 (011) give descending 110 minus ascending 011 = 099, not themselves - so they are not fixed. 100 gives $100 - 001 = 099$, changing. Multiples of 9 are not generally fixed (the routine's outputs are multiples of 9, but being a multiple of 9 does not make a number fixed). Only 495 is fixed, and it exceeds 100.

45. (A) Reflecting left-to-right reverses each row: every row keeps the same three numbers, so its sum is unchanged (still 15). The three columns are simply relabelled in reverse order, so each column sum still equals one of the original column sums (15). The two diagonals swap with each other under the reflection, and since both originally summed to 15, both still do. Hence the reflected grid is again magic. The diagonals are not destroyed - they exchange; the centre stays put (it maps to itself under a horizontal flip); and the property holds for any magic square, not only the 1-9 one, though here it does use 1-9.

46. (C) The Collatz conjecture - verified for all numbers far beyond 27 - states that the process always reaches 1, and 27 is a well-known case that climbs very high (peaking over 9000) but still descends to 1 after 111 steps. So we can be certain it ends at 1. It does not grow forever (it is known to terminate at 1 for 27); 6174 is Kaprekar's constant, a different process; and the process does not 'end' at its peak - the peak is just the highest value along the way before it falls to 1.

47. (C) Let the number be $10t + u$. We need $10t + u = 7(t + u)$, so $10t + u = 7t + 7u$, giving $3t = 6u$, hence $t = 2u$. The digit pairs with $t = 2u$ are $u=1, t=2$ (21); $u=2, t=4$ (42); $u=3, t=6$ (63); $u=4, t=8$ (84). All four satisfy the equation (for example $21 = 7 \times 3$ ✓ and $42 = 7 \times 6$ ✓), so the smallest of them is 21. The values 42 and 63 are larger valid solutions, not the smallest; 14 fails the rule ($7 \times (1+4) = 35$, not 14).

48. (C) The largest a single row can sum to is $7 + 8 + 9 = 24$, using the three biggest numbers in one row. The extra rule only requires 1 to sit somewhere on the main diagonal; we can place 1 on the diagonal in a DIFFERENT row from the 7-8-9 row, leaving 7,8,9 free

to share a row. For example, put 7,8,9 in the top row and 1 on the diagonal in the middle or bottom position - fully compatible. So the maximum row sum remains 24. The value 23 wrongly assumes 1 must displace a big number; 22 assumes two displacements; 15 confuses this with the magic-square row sum.

49. (B) Each step multiplies by roughly 1.618, so three steps (from 7th to 10th) multiply by about 1.618^3 . Compute $1.618^2 \approx 2.618$, then $\times 1.618 \approx 4.24$. So the 10th term is about 4.2 times the 7th. Checking exactly: 7th = 13, 10th = 55, and $55/13 \approx 4.23$ ✓. The value 1.6 is just one step; 3 underestimates 1.618^3 ; 8 confuses it with 2^3 (treating the ratio as 2).

50. (C) A 4-digit palindrome in 2000-2999 has the form 2 b b 2 (first digit 2 forces last digit 2; the middle two must match). The middle digit b runs 0 through 9, giving 2002, 2112, 2222, 2332, 2442, 2552, 2662, 2772, 2882, 2992 - that is 10 years. The value 9 forgets b=0; 11 adds an extra; 100 wrongly treats both middle digits as free.

51. (B) A call at position p must satisfy $0 \leq \text{call} \leq p - 1$. Positions 1-6 allow maxima 0,1,2,3,4,5. The calls 0,1,1,2,3,5 give: position1=0 (≤ 0 ✓), position2=1 (≤ 1 ✓), position3=1 (≤ 2 ✓), position4=2 (≤ 3 ✓), position5=3 (≤ 4 ✓), position6=5 (≤ 5 ✓). Every call is within its bound, and one can construct heights realising them, so the sequence is valid. Repeated values like two 1's are allowed (two different children can each have one taller person ahead); the last call 5 is exactly the maximum for position 6, hence fine; the heights need not follow any Fibonacci pattern - that the CALLS resemble Fibonacci is a coincidence.

52. (A) Palindromes of the form a0a with a from 1-9 are 101, 202, 303, ..., 909. Each equals $100a + a = 101a$. Their sum is $101 \times (1 + 2 + \dots + 9) = 101 \times 45 = 4545$. The value 4995 mistakenly uses 111×45 (treating the number as a repdigit aaa rather than a0a); 5050 and 4950 are arithmetic slips.

53. (D) Each row has 3 cells. Adding 10 to every cell adds $3 \times 10 = 30$ to each row's total. The new magic sum is $15 + 30 = 45$. It is still a magic square (every line gains the same 30), with magic sum 45. The value 25 adds only 10 (one cell's worth); 15 forgets the shift entirely; 150 multiplies instead of adds.

54. (D) From 16: $16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$, which is 4 halving steps. Because $16 = 2^4$ is a power of 2, the process never needs the $3n+1$ branch - it just halves four times. The count 5 overcounts; 'multiple of 4' is true but does not capture why it is so short (8 and 4 are also multiples of 4 with fewer steps - the key is being a pure power of 2); '16 steps' has no basis.

55. (D) If a 3-digit number equals its own reversal, then reversing leaves it unchanged, which is precisely the definition of a palindrome - its first and last digits match (the middle stays put). It need not be a multiple of 11 (e.g. $121 = 11^2$ is, but $232 = 8 \times 29$ is a palindrome not divisible by 11); it cannot end in 0 (then it could not start with 0 after reversal and stay 3-digit); and it is 3-digit by assumption, not single-digit.

56. (C) Only the front child has nobody ahead, so its call is forced to 0. Every later child has at least one person ahead and COULD have a taller person there, so could call a positive number (e.g. the second child calls 1 if the front child is taller). Thus exactly one child - the first - is guaranteed unable to exceed 0. 'All of them' is false (later children often call positive numbers); the second child is not forced to 0; and the first child certainly cannot exceed 0 since nobody stands ahead.

57. (A) Two supercells cannot share an edge (each would need to beat the other). So the cell with 8 must not be edge-adjacent to the cell with 9. The opposite corner shares no edge with 9's corner, so 8 there can exceed its two (smaller) neighbours and be a genuine supercell while 9 stays one too. Placing 8 directly beside 9 makes them edge-adjacent, so 8 would lose to 9 and fail. The centre touches the maximal number of cells (4), making it hardest to isolate and likely adjacent to 9. And 8 certainly CAN be a supercell when its neighbours are all smaller, so 'nowhere' is false.

58. (A) Halving an even number gives a 3-digit result when the half is at least 100, i.e. the even number is at least 200, and we want the LARGEST 3-digit even number, since odd numbers do not halve. The largest even 3-digit number is 998 (999 is odd, so it would go to $3 \times 999 + 1 = 2998$, a 4-digit jump, not a halving). Halving 998 gives 499, still 3 digits. So 998 is the answer. 999 is odd and does not halve; 500 is even and halves to 250 but is not the largest; the last option restates 999 incorrectly.

59. (D) In such a sequence each term is the sum of the two before, so the 10th term = 9th + 8th. Thus $55 = 9\text{th} + 21$, giving $9\text{th} = 55 - 21 = 34$. Check: 21, 34, 55 - indeed $21 + 34 = 55$ ✓. The value 33 mis-subtracts; 38 adds wrongly; 29 averages instead of subtracting.

60. (C) A 3-digit palindrome aba is even when its last digit (which equals a) is even, so a is even. The even choices for a (the leading digit, 1-9) are 2, 4, 6, 8 - that is 4 values. The middle digit b is free over 0-9 - 10 values. So there are $4 \times 10 = 40$ even 3-digit palindromes. The value 45 wrongly uses 9 choices for a (all leading digits) $\times \dots$; 50 uses 5 even digits including 0 as a leading digit (not allowed); 20 halves incorrectly.